

NOTE: We would like to record this meeting to make it available to those that were not able to attend or for you to review later. Please raise concerns or objections if you do not want this session recorded.

Southeast Texas Hurricane Evacuation Study

Transportation Analysis Kick-Off Meeting

October 28, 2024



Agenda:

- Re-Introductions to the Team
- Overview of HES progress
 - Evacuation zone development
 - Atlas updates and vulnerability layers
- Transportation Analysis
 - Overview
 - Scenario development
- Next Steps
 - Regional meetings
- Questions and Discussion
- Contacts

Project website:



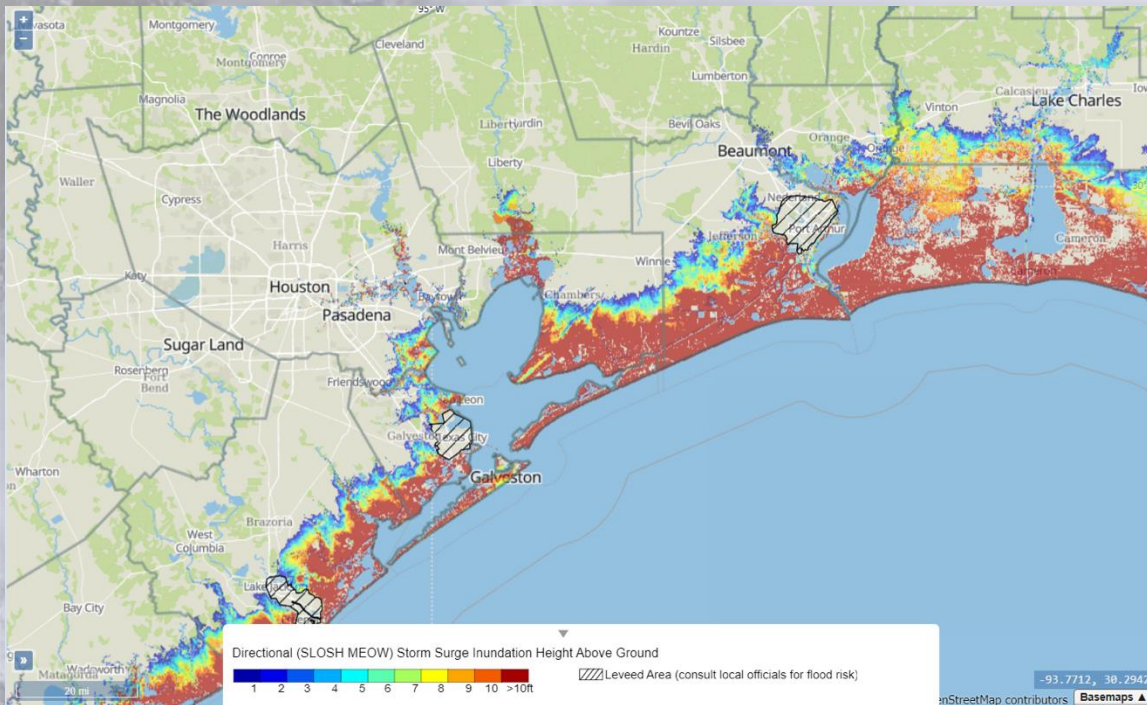
Team Members:

- US Army Corps of Engineers: Galveston District
 - Kyle Donlevy & Gerald Gaines
- FEMA, Region 6
 - Arianne Thomas
- Texas Department of Emergency Management
 - Blake White & Carman Apple
- Texas A&M Hazard Reduction and Recovery Center & Texas A&M Transportation Institute
 - Walt Peacock, David Bierling, Doug Wunneburger, Darrell Borchardt, & Alexander Abuabara
- Local government and stakeholder
 - County Judges, Local Emergency management organizations, and other stakeholders
 - **Please put your name, organization, county, and contact information in the chat**

HAZARDS ANALYSIS

SLOSH Data- Review

Cat 3, NW, 15 mph



- NE SLOSH basin
- Coastal features
- Bathymetry

SLOSH

MEOWs

- Direction
- Track
- Category
- Forward Speed

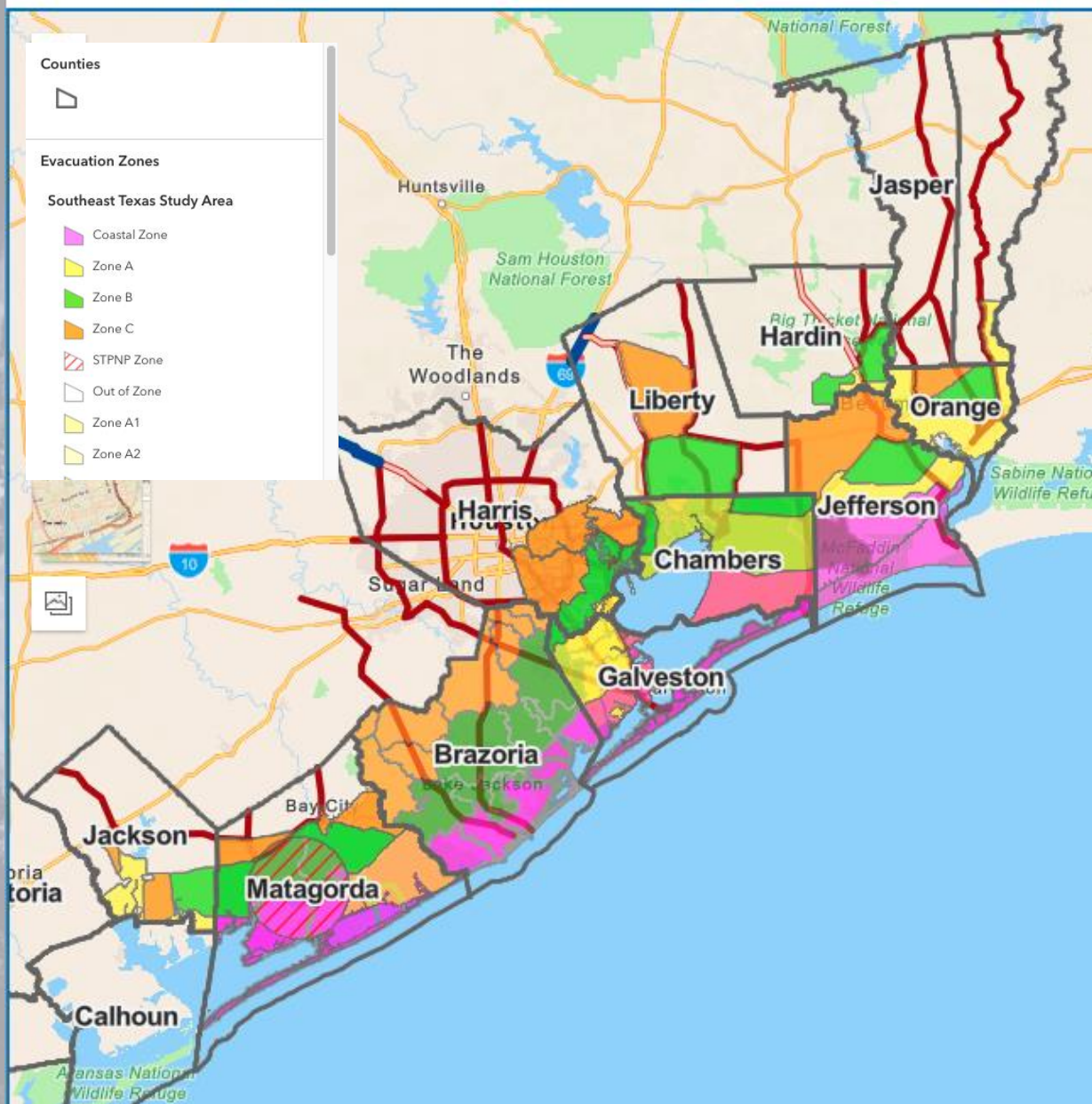
- Max storm tides
- All scenarios

MOMs

- **Total of 216 MEOWs**
 - 9 Directions (WNW through Parallel)
 - 6 Intensities: TS-5
 - 4 Forward Speeds
 - One tide (high)
- **Total of 6 MOMs**
 - 6 Intensities: TS-5



Southeast Texas Hurricane Evacuation Study



Update on Evacuation Zone reassessment and development

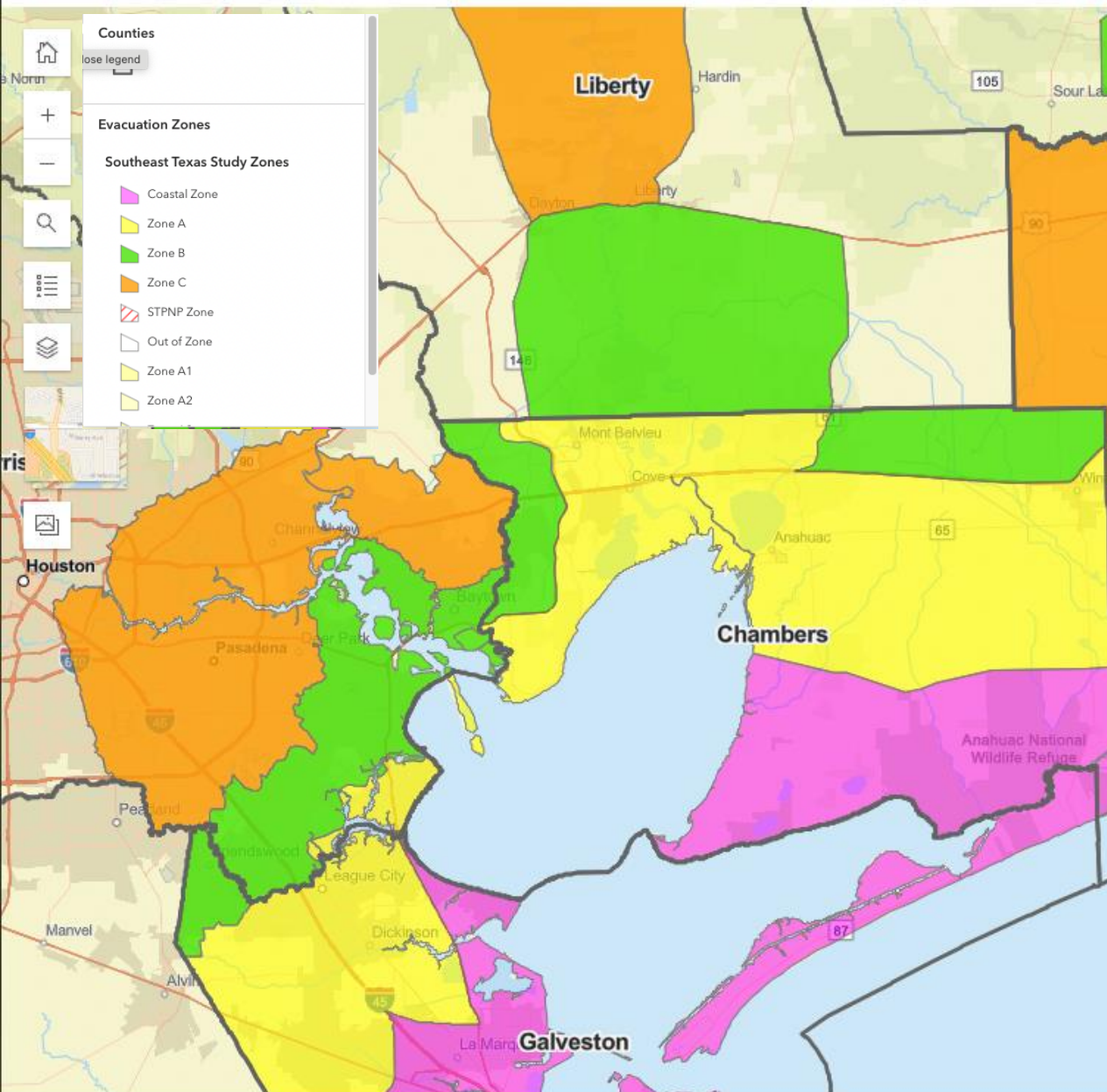
- The process is nearly complete
 - Few more tweaks are underway
 - Please, continue to review, share, and discuss with County Judges, and get with us if there are additional modifications.
- Atlas includes
 - Earlier evacuation zones
 - Layer name: **2023 HGAC Zip Zone**
 - The new draft zones
 - Layer name: **Southeast Texas Study Zones**
 - Change layers for areas increased, decreased, or remained unchanged
 - Layer name: **Change in Zone from 2023 HGAC Study**



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Southeast Texas Hurricane Evacuation Study

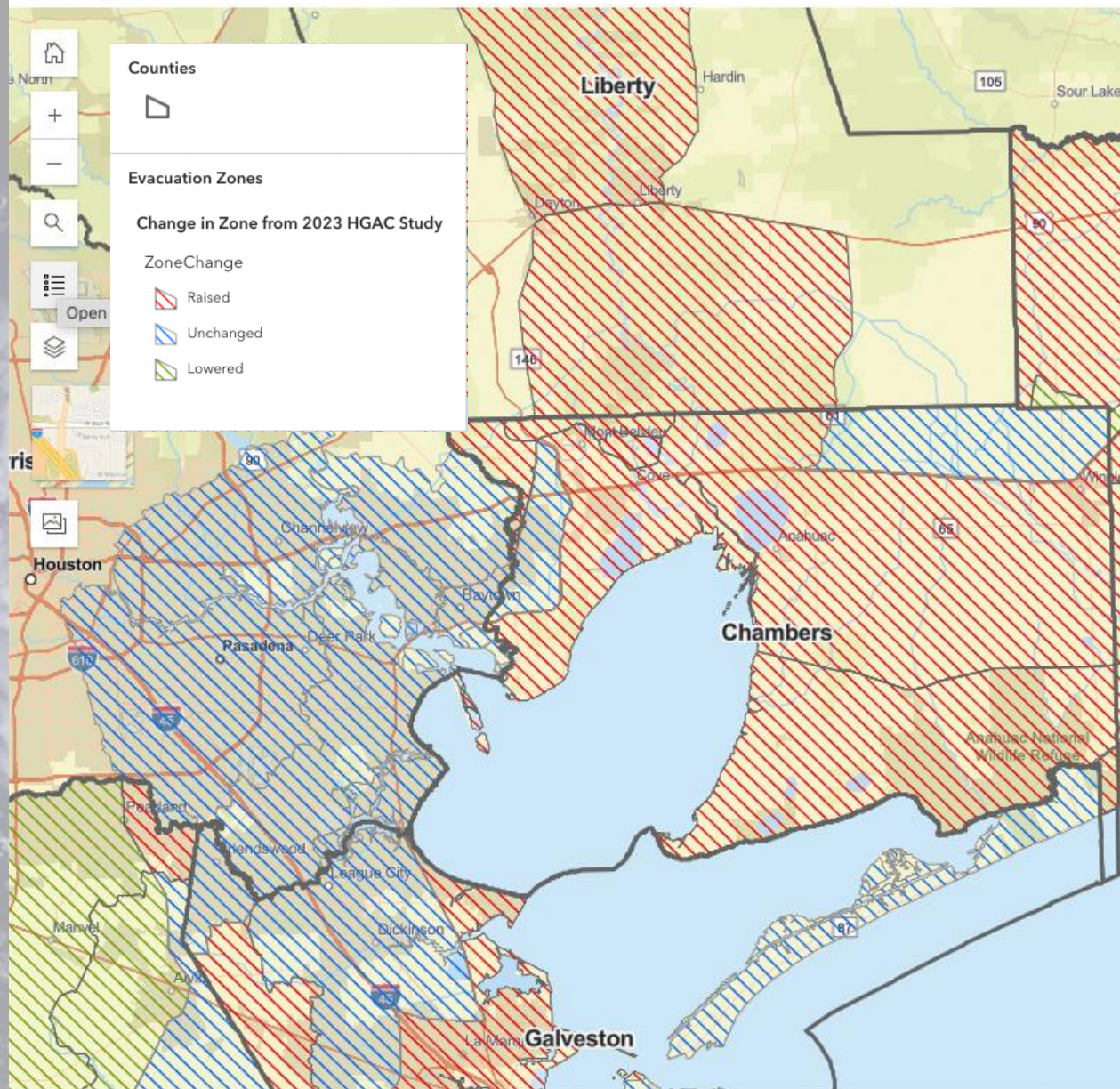


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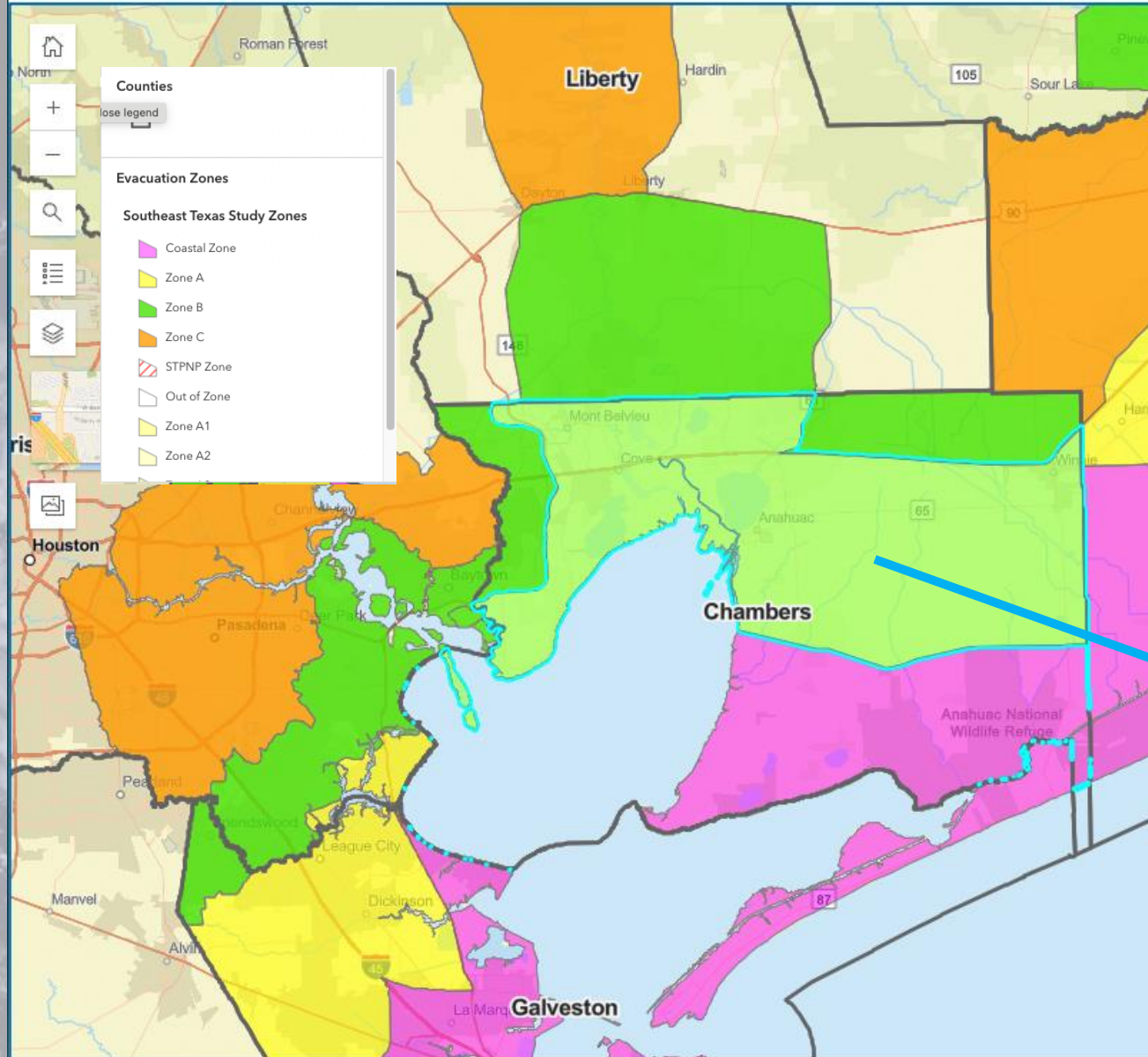


Southeast Texas Hurricane Evacuation Study



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Vulnerability data has been added and much more to come

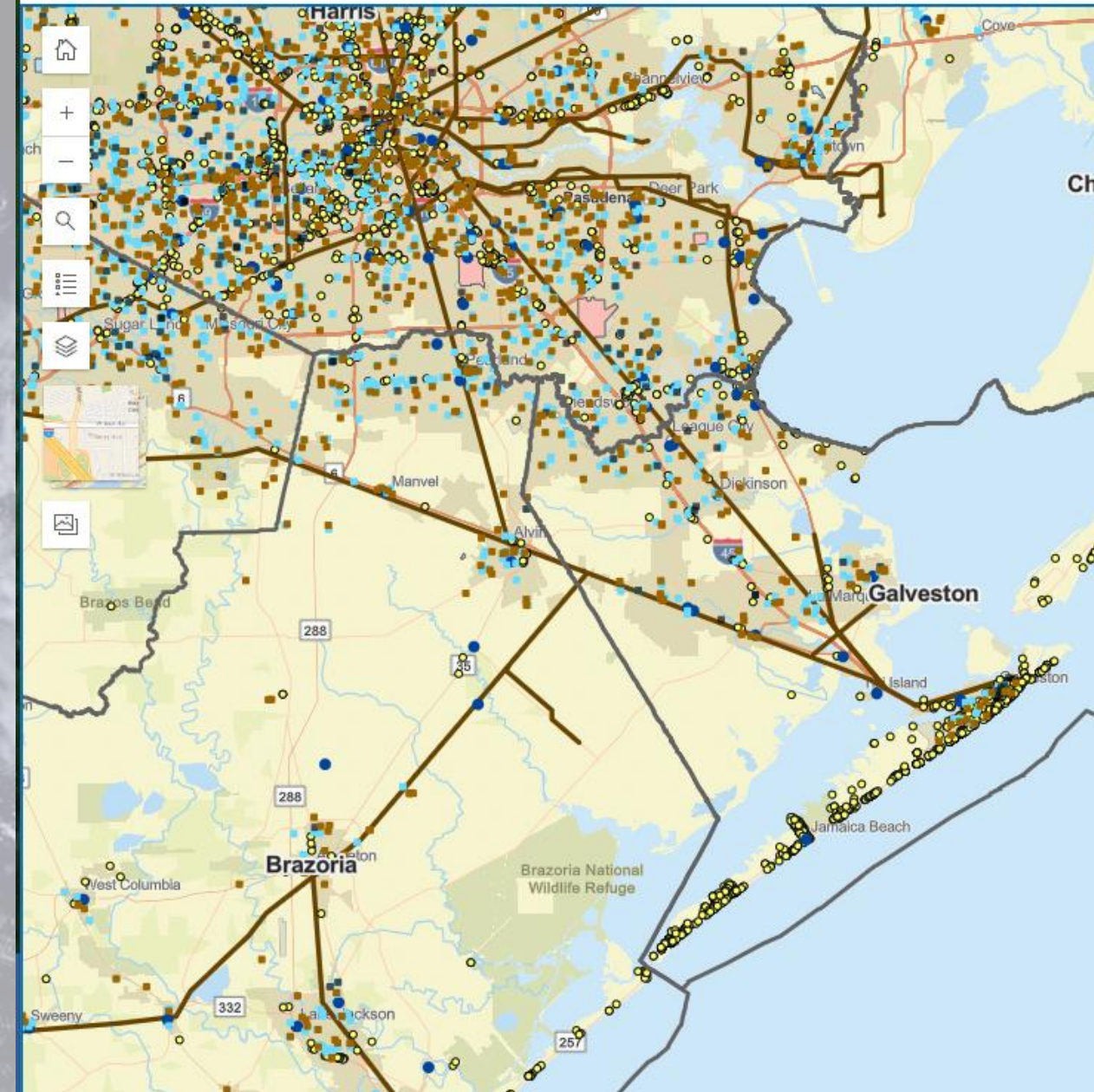
- Click on any County's evacuation zone and the following data will appear:
 - Population
 - Households
 - Vehicles
 - Zone's Boundary description

Chambers County Zone A

Zoom to

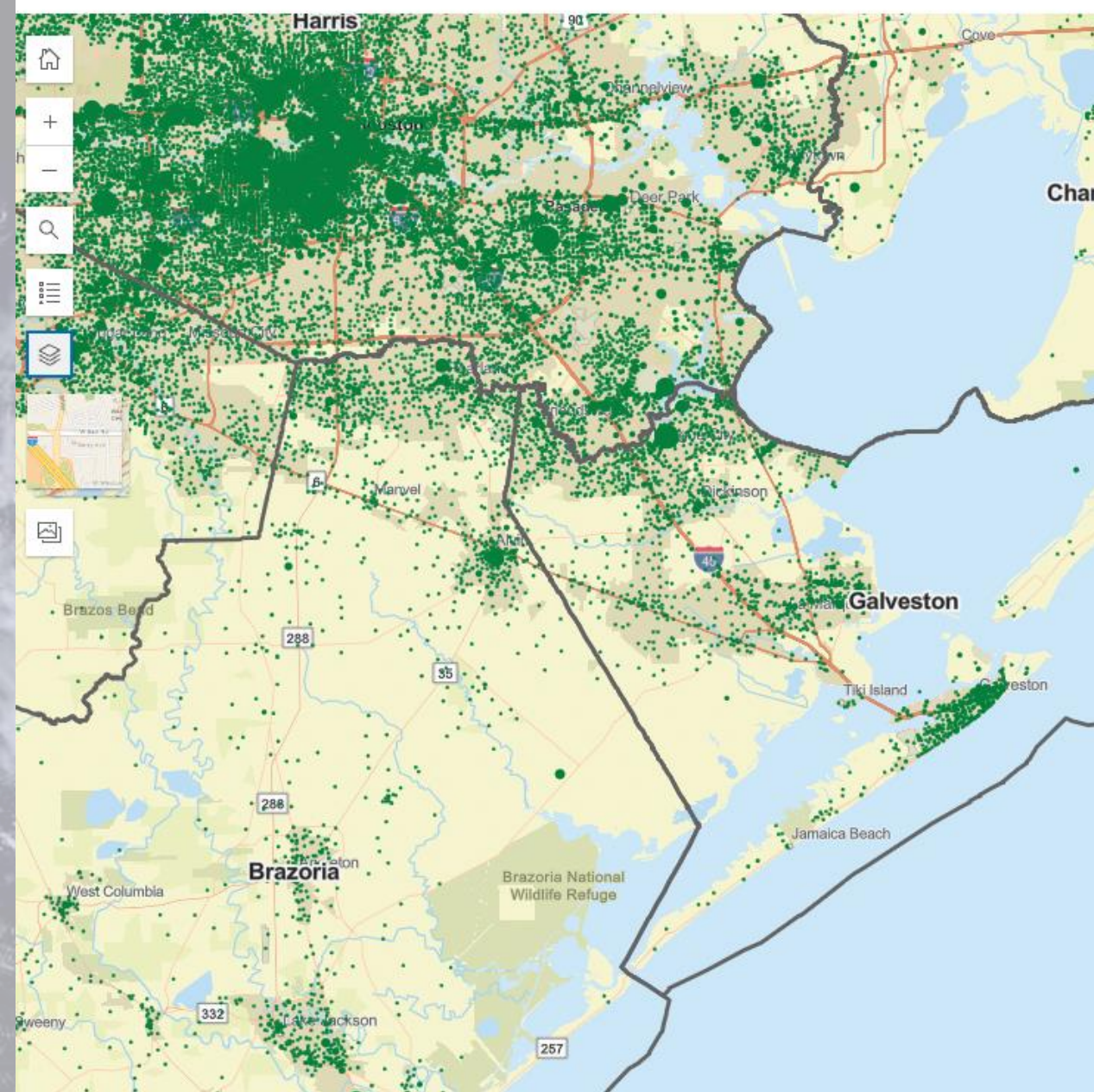
Population, Household, and Vehicle Estimates

Population (night)	33,194
Households (est)	11,284
Vehicles (est)	25,222
Zone Boundary	All areas east of Grand Pkwy to SH 61. South of IH 10.



Vulnerability data has been added and much more to come

- Built environment data
 - Hotels/motels/B&Bs
 - Childcare Centers
 - Schools (private and public)
 - Evacuation routes
 - Etc.
- Population concentrations and characteristics
 - Population concentrations
 - Social vulnerability indicators
 - Low to moderate income, etc.
 - Block group, tract, and ZCTAs
- Additional hazard layers
 - Surge, flood, levee areas,



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SETX HES Transportation Analysis

The primary purpose of the transportation analysis is to estimate the clearance times needed to complete a safe and timely evacuation of the population for a range of hurricane threats and scenarios.

- Identify Evacuation Network
- ***Determine Evacuation Scenarios***
- *Estimate Clearance Times*

What is the END GAME?

State: Texas ▼ County: Nueces ▼ Use Base Location

HURREVAC makes recommendations for evacuation start times based on how long it takes to evacuate a vulnerable population ahead of the arrival of tropical-storm-force winds (34kt/39mph). To utilize this capability of the program, you must first select one or more evacuation scenarios from a region's Hurricane Evacuation Study. Refer to the Study's technical data report, or ask your state's Hurricane Program Manager for guidance on making selections appropriate to a particular storm situation.

Total Evacuation hours: 41

Evacuation Participation Rate: Normal participation based on recent experience ▼

Response: Moderate (24 hour) response ▼

Roadway Modification: Additional capacity by utilizing breakdown lane on IH-37 ▼

Total Evacuation hours: 47

Evacuation Participation Rate: Normal participation based on recent experience ▼

Response: Moderate (24 hour) response ▼

Roadway Modification: No change to transportation system ▼

Total Evacuation hours: 36

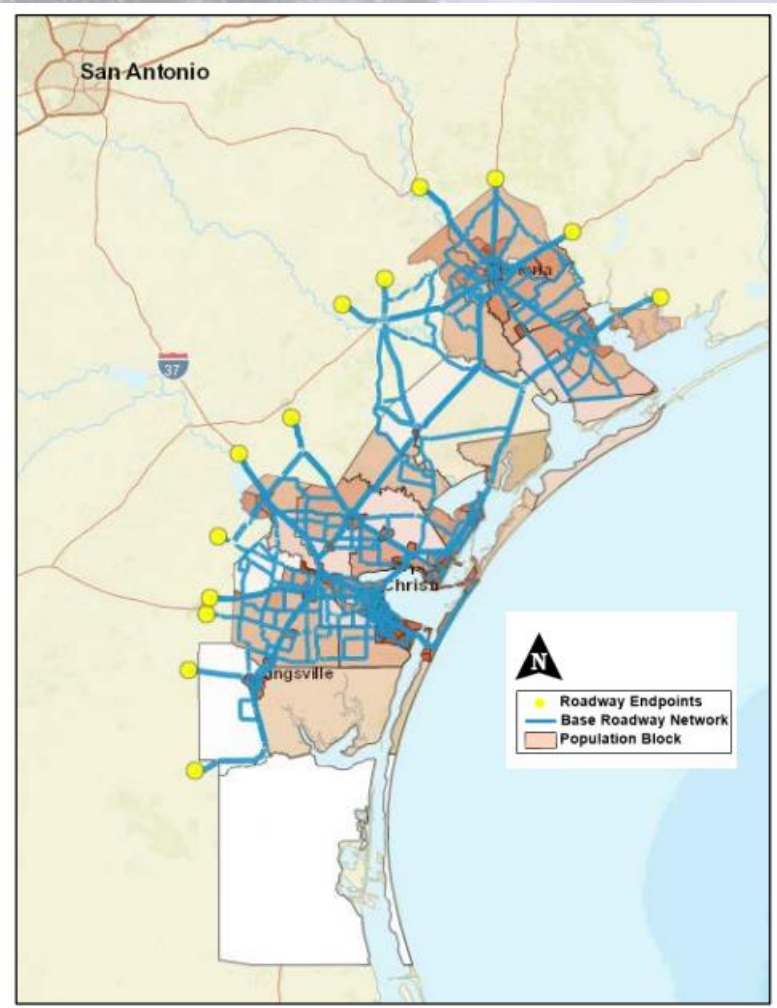
Evacuation Participation Rate: Normal participation based on recent experience ▼

Response: Moderate (24 hour) response ▼

Roadway Modification: Lane reversal implemented on IH-37 ▼

How do we get there?

Through the magic of RtePM!!



RtePM → Real time evacuation Planning Model

- **Model includes the existing street network**
 - ***Allows for modifications, closures, shoulder lanes, contraflow**
- **Population totals from latest census data**
 - **Daytime & Nighttime**
 - **Seasonal populations can be added**
- **Start time of evacuation can be staggered amongst zones**
- **Behavioral analysis used to help define evacuating traffic**
- **RtePM result = estimated time for the last car to reach endpoint**
 - **NOT all the way to Dallas**
 - **Reasonable safe location**

Scenario Development

Scenario Inputs

- Resident Behavior
 - Response Rate
 - Response Time
 - Destinations
- Seasonal Population
- Roadway Operations
 - Evaculane
 - Contraflow
 - Incidents
 - Background Traffic
- “All Scenario” Inputs
 - Percent private vehicles
 - Percent using other means
 - Average Vehicle Occupancy
 - Vehicles Towing

Scenario Development

Resident Behavior

- Storm impact and timing
- Behavior study
- Rely on local knowledge/expertise
- Possible vs. Practical # of scenarios

Response Time

- Fast (5 hours)
- Medium(24 hours)
- Slow (2-day)
- Custom

Scenario Development

Roadway Operations

- **Road Closures**
- **Contraflow**
- **Evaculane**
- Traffic incidents
- Background traffic

Scenario Development

NOTE: Charts showcasing scenarios used for Coastal Bend study for illustrative purposes only. NEW scenarios will be developed specifically for the 2024 Southeast Texas study.

Scenario Group	Population	Evacuation Zone/Area				Persons Evacuating
		A	B	C	Outside Evacuation Zone	
Recent Experience	% Resident Pop	75%	51%	40%	44%	377,000
	% Seasonal Pop	75%	51%	40%	44%	
Recent Experience Plus	% Resident Pop	90%	66%	55%	59%	479,200
	% Seasonal Pop	90%	66%	55%	59%	
Maximum Evacuation	% Resident Pop	100%	100%	100%	100%	681,400
	% Seasonal Pop	100%	100%	100%	100%	

Scenario Group	Evaculane	Contraflow	Response Time
Recent Experience			2 days
Scenario Group	Evaculane	Contraflow	Response Time
Recent Experience Plus			
Scenario Group	Evaculane	Contraflow	Response Time
Maximum Evacuation	No	No	2 days 24 hours 5 hours
	Yes	No	2 days 24 hours 5 hours
	Yes	Yes	2 days 24 hours 5 hours



TRANSPORTATION ANALYSIS

Outcomes



- Develop Evacuation Scenarios through close coordination with local & state emergency managers
- Transportation modeling to get Clearance Times using the Real Time Evacuation Planning Model (RtePM)
- HURREVAC Integration

Scenario	Subregion					Dir			Cat			Zones				Part Rate			LR		NC		Likely		REGIONAL CLEARANCE TIME
	SS	P	MP	NN	ES	NW	NE	1/2	3	4	A	B	C	D	L	M	H	W	WO	W	WO	ML	CAT		
ES-1																									28
ES-2																									28
ES-3																									45
ES-4																									38
ES-5																									58
ES-6																									50
ES-7																									34
ES-8																									45
ES-9																									38
ES-10																									58
ES-11																									58
ES-12																									50
ES-13																									67
ES-14																									67
ES-15																									58



Evacuation Scenarios | Timeline Actions | Timing Arcs

State: County:

HURREVAC makes recommendations for evacuation start times based on how long it takes to evacuate a vulnerable population ahead of the arrival of tropical storm force winds (34kt/39mph). To utilize this capability of the program, you must first select one or more evacuation scenarios from a region's Hurricane Evacuation Study. Refer to the Study's technical data report, or ask your state's Hurricane Program Manager for guidance on making selections appropriate to a particular storm situation.

Virginia HES 2020

Total Evacuation hours: Range of 15 hours - 96 hours

Internal Regions Evacuating:

External Regions Evacuating:

Evacuation Zone:

Storm Direction:

Evacuation Participation Rate:

Roadway Modifications:

Saved Scenarios | Criteria Selection

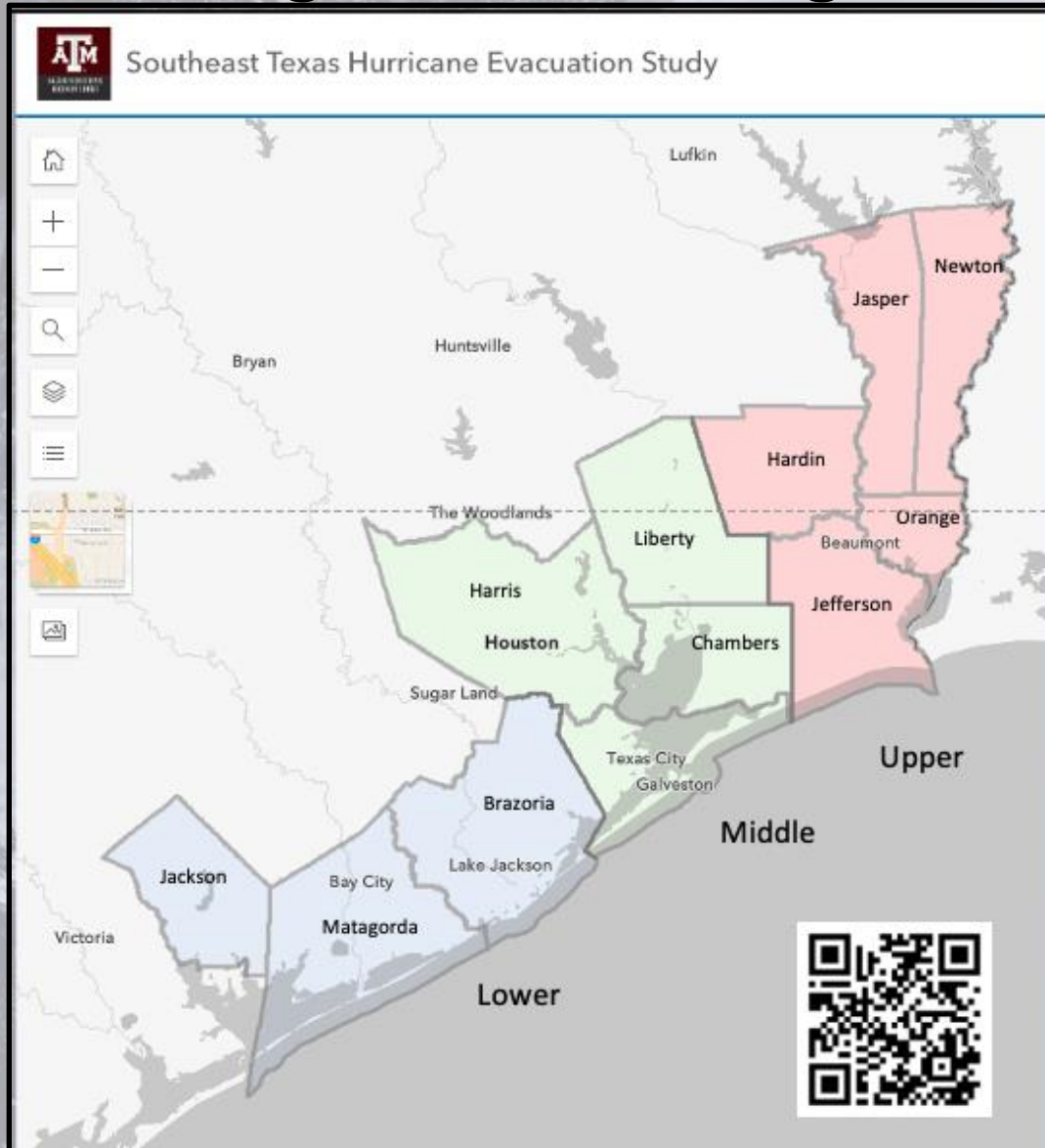
☐ Location	Scenario	Hours
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Transportation Analysis / Scenario Development

Scenarios “drive” the transportation analysis

- ✓ Need your input to develop scenarios [each region can be different]
 - Staggered zone start times
 - Road Closures
 - Evaculane/Contraflow options
 - Seasonal population
 - Special needs population not in private vehicles
 - Population to shelters
- ✓ Resident behavior
 - Response time
 - Likely destinations
- ✓ RtePM variables to consider
 - Persons per vehicle
 - % vehicles towing
 - Background traffic / roadway incidents

Transportation Analysis: First Regional Meetings



- **Meeting #1 Middle Planning Zone**
 - Dates: 11/18/24, 8AM to 11AM
 - Chambers, Galveston, Harris, and Liberty Counties
 - Location: Walter Hall Park , 807 State Hwy 3 N.
 - League City TX 77573
- **Meeting #2 Lower Planning Zone**
 - Date: 11/18/24, 1PM to 4PM
 - Brazoria, Jackson, and Matagorda Counties
 - Location: Brazosport College, Rm. 103AB in the Corporate Learning Center, 500 College Drive, Lake Jackson, 77566
- **Meeting #3 Upper Planning Zone**
 - Date: 12/2/24, 8AM to 11AM
 - Hardin, Jasper, Jefferson, Newton, and Orange Counties
 - Location: Orange County Convention and Expo Center, 11475 FM 1442, Orange, TX 77630

Agendas and additional information will be distributed prior to meetings.

Please put your name, organization, county, and contact information in the chat so we can contact you.

Atlas website:



Project website:



Questions & Discussion

Contacts



- USACE Galveston District

- **Kyle Donlevy:** kyle.a.donlevy@usace.army.mil
 - Overall, HES Re-Study Manager and coordination



- FEMA Region 6

- **Arianne Thomas:** arianne.deruise@fema.dhs.gov
 - HES input and technical support

- Texas Department of Emergency Management



- **Blake White:** blake.white@tdem.texas.gov
- **Carman Apple:** carman.apple@tdem.texas.gov
 - HES oversight, input and technical support, coordination with county, local, & regional government, agencies, and stakeholders.

- Texas A&M HRRC and TTI



- Conducting vulnerability, behavioral, shelter, and transportation analysis and providing technical assistance.
 - **Walt Peacock:** peacock@tamu.edu
 - Overall team management, coordination, and data analysis
 - **David Bierling:** d-bierling@tti.tamu.edu
 - Overall team management, coordination, and data analysis
 - **Doug Wunneburger:** dwunneburger@arch.tamu.edu
 - GIS and data development and analysis, website development
 - **Darrell Borchardt:** d-borchardt@tti.tamu.edu
 - Transportation scenario development and analysis
 - **Alexander Abuabara:** aabuabara@arch.tamu.edu
 - GIS & data development and analysis and website development and maintenance